

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	Course Title:	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	Duration / Marks:	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	AI & DS	Date:	05/10/2026

Code of Conduct

I, B. Muni Chandini (Reg. No: 192524283), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Muni Chandini

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evaluation page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	Q3
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	Q1, Q2
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	Q1, Q4

Annexure A – Analytical Problem Solving

Question 1: Two-Room Vacuum World

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of $T = 6$ time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take $p = 1$ unless told otherwise).

- Agent X (simple reflex): if the current square is Dirty, Suck; otherwise move to the other square.
 - Agent Y (reflex with internal state): if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.
- (a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. [4 Marks]
- (b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for $p = 1$. [5 Marks]
- (c) Trace Agent Y in the same way and find its total score for $p = 1$. [4 Marks]
- (d) Derive the total score of X and of Y as functions of p . For which values of p is Y strictly better than X? [3 Marks]
- (e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. [4 Marks]

Question 2: Task Environment Analysis and Agent Selection

Three AI agents are to be deployed in the following task environments:

- E1: An autonomous taxi driving through city traffic.
- E2: A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- E3: A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

- (a) Classify each environment on all six properties with a one-phrase justification for each. [9 Marks]
- (b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. [4 Marks]
- (c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. [4 Marks]
- (d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. [3 Marks]

Question 3: Turing Test Data Across the History of AI

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

- (a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. [6 Marks]
 - (b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. [5 Marks]
 - (c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. [5 Marks]
 - (d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). [4 Marks]
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Question 4: Rational Route Choice under Uncertainty

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where T is the travel time in minutes and $L = 1$ if the taxi arrives after the 55-minute deadline, otherwise $L = 0$.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

- (a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. [4 Marks]
- (b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. [5 Marks]
- (c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. [4 Marks]
- (d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. [4 Marks]
- (e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. [3 Marks]
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ANSWERS

ANSWER – Question 1: Two-Room Vacuum World

(a) PEAS Description

Method: Extract each PEAS component directly from the problem data.

Component	Description
Performance	+1 for every clean square at the end of each time step; penalty of $-p$ for every Left or Right move. Lifetime $T = 6$ steps.
Environment	Two squares (A and B). Both initially Dirty. Dirt never reappears once cleaned. Agent starts in A. Fully observable, deterministic, sequential, static, discrete, single-agent.
Actuators	Suck (clean current square), Right (move to B), Left (move to A), NoOp (do nothing).
Sensors	Location sensor (A or B) and dirt sensor (Dirty or Clean) for the current square only.

(b) Trace of Agent X (Simple Reflex) for $p = 1$

Agent X rule: If current square is Dirty $\rightarrow$ Suck; otherwise $\rightarrow$ move to the other square.

Assumptions: Start at A, both Dirty. After Suck, square becomes Clean permanently. Move incurs penalty $p = 1$. Step score = (number of clean squares) – (move penalty if any).

Step	Loc	Percept	Action	A	B	Clean#	Score
1	A	A, Dirty	Suck	Clean	Dirty	1	$1 - 0 = 1$
2	A	A, Clean	Right	Clean	Dirty	1	$1 - 1 = 0$
3	B	B, Dirty	Suck	Clean	Clean	2	$2 - 0 = 2$
4	B	B, Clean	Left	Clean	Clean	2	$2 - 1 = 1$
5	A	A, Clean	Right	Clean	Clean	2	$2 - 1 = 1$
6	B	B, Clean	Left	Clean	Clean	2	$2 - 1 = 1$

Total score of Agent X ($p = 1$) = $1 + 0 + 2 + 1 + 1 + 1 = 6$

Working: After both rooms are cleaned (step 3), Agent X has no memory and continues oscillating Left/Right, paying a move penalty every step while scoring 2 for clean squares, netting 1 per step.

(c) Trace of Agent Y (Reflex with Internal State) for $p = 1$

Agent Y rule: If current square Dirty $\rightarrow$ Suck; else if it remembers both squares cleaned $\rightarrow$ NoOp; else $\rightarrow$ move to the other square.

Step	Loc	Percept	Action	A	B	Clean#	Score
1	A	A, Dirty	Suck	Clean	Dirty	1	$1 - 0 = 1$
2	A	A, Clean	Right	Clean	Dirty	1	$1 - 1 = 0$
3	B	B, Dirty	Suck	Clean	Clean	2	$2 - 0 = 2$
4	B	B, Clean	NoOp	Clean	Clean	2	$2 - 0 = 2$
5	B	B, Clean	NoOp	Clean	Clean	2	$2 - 0 = 2$
6	B	B, Clean	NoOp	Clean	Clean	2	$2 - 0 = 2$

Total score of Agent Y ($p = 1$) = $1 + 0 + 2 + 2 + 2 + 2 = 9$

Working: After cleaning both rooms (step 3), Agent Y's internal state records the world is fully clean and issues NoOp thereafter, avoiding move penalties.

(d) Scores as Functions of p; When is Y Strictly Better?

Agent X performs 4 moves (steps 2, 4, 5, 6). Clean-square total = 10. Move cost = 4p.

$$\text{Score_X}(p) = 10 - 4p$$

Agent Y performs 1 move (step 2). Clean-square total = 10. Move cost = 1p.

$$\text{Score_Y}(p) = 10 - p$$

Y is strictly better when $10 - p > 10 - 4p \Rightarrow 3p > 0 \Rightarrow p > 0$.

For all $p > 0$, Agent Y is strictly better than Agent X.

(e) Which Agent is Rational? Interpretation

Agent Y is the rational agent.

A rational agent selects the action that maximises expected performance measure given the percept sequence to date. After both rooms are cleaned, Y's internal state (summarising the percept history) tells it that further movement yields only cost; hence NoOp is optimal.

Agent X cannot match Y even though both receive the same external percepts, because X is a pure simple-reflex agent whose action depends only on the current percept. It has no memory of history, so it never "knows" the other room is already clean and keeps oscillating. Rationality therefore depends on both the performance measure and the ability to exploit the entire percept history (or an internal model that summarises it).

ANSWER – Question 2: Task Environment Analysis and Agent Selection

(a) Classification of Each Environment

E1 – Autonomous Taxi Driving through City Traffic

Property	Value	Justification
Observability	Partial (1)	Cannot sense all vehicles, pedestrians, or road conditions beyond sensor range.
No. of agents	Multi (1)	Other drivers, pedestrians, traffic systems act as independent agents.
Determinism	Stochastic (1)	Sudden braking, weather, unpredictable human behaviour introduce uncertainty.
Episodic/Seq.	Sequential (1)	Current decisions (route, speed, lane) affect future states.
Static/Dynamic	Dynamic (1)	Environment changes continuously while the agent deliberates.
Discrete/Cont.	Continuous (1)	Speed, position, steering angle are continuous variables.

E2 – Part-Picking Robot on Moving Conveyor Belt

Property	Value	Justification
Observability	Partial (1)	Camera sees only parts in its field of view; may miss occluded defects.
No. of agents	Single (0)	Only one robot acts; conveyor is part of the environment.
Determinism	Stochastic (1)	Part orientation, lighting, gripper slip introduce uncertainty.
Episodic/Seq.	Episodic (0)	Each part-picking decision is largely independent of previous picks.
Static/Dynamic	Semi-dyn (0.5)	Belt moves, but deliberation time is short relative to belt speed.
Discrete/Cont.	Continuous (1)	Arm joint angles and part positions are continuous.

E3 – Chess-Playing Agent in Tournament with Clock

Property	Value	Justification
Observability	Fully (0)	Entire board state and remaining clock times are visible.
No. of agents	Multi (1)	Two opposing players; each is an independent agent.
Determinism	Deterministic (0)	Rules of chess are fully deterministic.
Episodic/Seq.	Sequential (1)	Each move alters the board; future options depend on game history.
Static/Dynamic	Semi-dyn (0.5)	Board is static during thought, but the clock runs continuously.
Discrete/Cont.	Discrete (0)	Finite set of legal moves; board positions are discrete.

(b) Difficulty Scores and Ranking

Env	Obs	Agents	Det	Epi/Seq	Stat/Dyn	Disc/Cont	Total
E1 (Taxi)	1	1	1	1	1	1	6.0
E2 (Robot)	1	0	1	0	0.5	1	3.5
E3 (Chess)	0	1	0	1	0.5	0	2.5

Ranking (hardest → easiest): E1 (Taxi, 6.0) > E2 (Robot, 3.5) > E3 (Chess, 2.5)

(c) Suitable Agent Architectures

Hardest – E1 (Autonomous Taxi): Utility-based agent (or Learning agent).

Justification: Partially observable, multi-agent, stochastic, sequential, dynamic and continuous. A utility-based agent can maximise a multi-attribute utility (safety, time, comfort, fuel). A learning agent can improve its model from experience.

Easiest – E3 (Chess): Goal-based agent (adversarial search).

Justification: Fully observable, deterministic, discrete and only semi-dynamic. A goal-based agent performing minimax / alpha-beta search toward checkmate is classical and sufficient.

(d) Interpretation

The domain label (taxi, robot, chess) is only surface description. What determines agent-program complexity are the six task-environment properties. A partially observable, stochastic, multi-agent, continuous, dynamic environment forces belief maintenance, uncertainty handling and multi-objective optimisation—hence utility-based or learning architectures. A fully observable, deterministic, discrete, single-agent, static environment can be solved by simple search. Thus environment properties, not domain names, decide how complex the agent must be.

ANSWER – Question 3: Turing Test Data Across the History of AI

(a) Correct-Identification Rate and Fooling Rate

Method: $\text{Correct-identification rate} = (\text{correctly identified} / 50) \times 100\%$. $\text{Fooling rate} = 100\% - \text{correct-identification rate}$.

System	Correctly ID'd	Correct Rate	Fooled	Fooling Rate
ELIZA	47 / 50	94%	3 / 50	6%
Chatbot B	36 / 50	72%	14 / 50	28%
System C	29 / 50	58%	21 / 50	42%

(b) Which Systems Satisfy Turing's 70% Criterion?

Turing (1950): an average interrogator has no more than a 70% chance of correct identification. Equivalently, correct-identification rate $\leq 70\%$ (fooling rate $\geq 30\%$).

Method: Compare each correct-identification rate with the 70% threshold; also measure distance from pure chance (50%).

System	Correct Rate	$\leq 70\%$?	Distance from 50% chance
ELIZA	94%	No	+44 percentage points above chance
Chatbot B	72%	No (just above)	+22 percentage points above chance
System C	58%	Yes	+8 percentage points above chance

Only System C satisfies Turing's 70% criterion (correct-identification rate $58\% \leq 70\%$).

(c) Improvement in Fooling Rate

ELIZA fooling rate = 6%; Chatbot B = 28%; System C = 42%.

Improvement ELIZA $\rightarrow$ System C:

- Absolute: $42\% - 6\% = 36$ percentage points
- Ratio: $42 / 6 = 7$ (System C fools judges 7 times more often than ELIZA)

Improvement Chatbot B $\rightarrow$ System C:

- Absolute: $42\% - 28\% = 14$ percentage points

(d) Interpretation: High Fooling Rate $\neq$ Thinking or Rationality

The Turing Test operationalises the "Acting Humanly" approach to defining AI. A high fooling rate shows only that the system can imitate human conversational behaviour well enough to deceive a judge for five minutes.

It does not prove that the system is thinking ("Thinking Humanly / Thinking Rationally") nor that it is acting rationally. Searle's Chinese Room argument illustrates this: a system can manipulate symbols by syntactic rules so skillfully that it passes a linguistic Turing Test, yet possess no genuine understanding or intentionality.

Likewise, a large language model may produce fluent text by statistical pattern matching without maintaining an explicit world model, goals or utility function—the hallmarks of rational agency. Thus "acting humanly" and "acting rationally" are distinct definitions; success on one does not entail success on the other.

ANSWER – Question 4: Rational Route Choice under Uncertainty

Utility: $U = 100 - T - 20 \cdot L$, where T = travel time (min), $L = 1$ if late ($T > 55$), else 0. Deadline = 55 min.

(a) Utility of Each Route in Each Outcome

Route	Outcome	T (min)	Late? (L)	$U = 100 - T - 20L$	Utility
R1 Highway	Normal	30	No ($L=0$)	$100 - 30 - 0$	70
R1 Highway	Disrupted	70	Yes ($L=1$)	$100 - 70 - 20$	10
R2 City	Normal	45	No ($L=0$)	$100 - 45 - 0$	55
R2 City	Disrupted	60	Yes ($L=1$)	$100 - 60 - 20$	20
R3 Ring	Always	50	No ($L=0$)	$100 - 50 - 0$	50

Late trips (clearly marked): R1 disrupted ($T = 70 > 55$) and R2 disrupted ($T = 60 > 55$).

(b) Expected Utility; Rational Choice

$$EU(R) = \sum P(\text{outcome}) \times U(\text{outcome})$$

$$EU(R1) = 0.70 \times 70 + 0.30 \times 10 = 49 + 3 = 52$$

$$EU(R2) = 0.90 \times 55 + 0.10 \times 20 = 49.5 + 2 = 51.5$$

$$EU(R3) = 1.00 \times 50 = 50$$

Rational agent chooses R1 (Highway) – highest expected utility ($52 > 51.5 > 50$).

(c) New Percept: $P(\text{disruption on R1}) = 0.80$

$$EU(R1') = 0.20 \times 70 + 0.80 \times 10 = 14 + 8 = 22$$

Comparison: $EU(R1') = 22$, $EU(R2) = 51.5$, $EU(R3) = 50$.

New rational choice: R2 (City road) with $EU = 51.5$.

(d) Actual Utility if Disruption Occurred; Rational vs Omniscient

Agent chose R1 in part (b). If disruption occurred: $T = 70$, $L = 1$.

Actual utility obtained = $100 - 70 - 20 = 10$.

Was the agent irrational? No. Rationality is defined with respect to expected utility given the information available at decision time, not the eventual outcome. At the moment of choice the agent correctly maximised EU under the then-known probabilities.

Distinction: A rational agent selects the action with highest expected performance given its percepts. An omniscient agent would know in advance which outcome will occur and could always pick the route yielding the highest actual utility. Omniscience is impossible in a stochastic world; rationality does not require it.

(e) Interpretation

Rationality is relative to (i) the performance measure ($U = 100 - T - 20L$) and (ii) the agent's current percept sequence / belief state. In part (b) the prior $P(\text{accident}) = 0.30$ made R1 optimal. The radio alert in part (c) is a new percept that revises the probability to 0.80; under the updated belief the same performance measure now ranks R2 highest. The physical roads did not change—only the information available to the agent changed. A rational agent maximises expected utility conditioned on the percepts it has received so far.